ReviewIQ AI: Customer Review Intelligence Platform

Emaad Faisal

1 Introduction

Customer reviews contain useful information about product quality, delivery, price, packaging, customer service, and customer satisfaction. However, businesses may receive a large number of reviews, making it difficult to read and analyze every review manually. ReviewIQ AI is an AI-powered customer review intelligence platform. It will use Natural Language Processing and machine learning to analyze customer reviews, classify sentiment, detect common complaints, and provide useful insights to businesses.

2 Problem Statement

Businesses receive customer feedback through e-commerce platforms, surveys, social media, and review websites. Manually analyzing a large number of reviews is slow and difficult. Star ratings may show whether customers are satisfied, but they do not fully explain the reasons behind customer satisfaction or dissatisfaction. For example, a customer may be satisfied with product quality but dissatisfied with delivery. Therefore, a single review may contain both positive and negative opinions. ReviewIQ AI aims to automatically analyze customer reviews and classify them as positive, neutral, or negative. The completed system will also identify common complaints and important aspects such as product quality, delivery, price, packaging, and customer service.

3 Target Audience

The target audience of ReviewIQ AI includes:

- E-commerce businesses
- Online product sellers
- Customer experience teams
- Product managers
- Marketing teams
- Businesses that collect customer feedback

The first version of the project will mainly focus on e-commerce customer reviews. The system may later be expanded for restaurants, hotels, service businesses, and other organizations.

4 AI Justification

Customer reviews are written in unstructured natural language. Different customers may express the same opinion using different words and writing styles. A review may also contain opinions about multiple aspects. For example:

> The product quality is excellent, but the delivery was late.

This review contains a positive opinion about product quality and a negative opinion about delivery. A basic rule-based program may detect both positive and negative words but may not understand which opinion belongs to each aspect. Machine learning and Natural Language Processing are suitable for this problem because they can learn language patterns from labelled customer reviews. A trained model can then classify previously unseen reviews instead of depending only on manually written rules. The project will use text preprocessing, TF-IDF feature extraction, and machine-learning classification models. Later, aspect detection will be added to identify the specific product or service feature discussed in each review.

5 Dataset Acquisition

The Amazon Reviews Multi English dataset was used for this project. The dataset was loaded from Hugging Face using the datasets library. The following Python code was used:

```
from datasets import load_dataset dataset = load_dataset( "SetFit/amazon_reviews_mul
```

The complete dataset contained 200,000 training reviews, 5,000 validation reviews, and 5,000 testing reviews. A smaller stratified sample was selected from each split so that the data could be processed more easily in Google Colab. The working sample initially contained:

- Training data: 30,000 reviews
- Validation data: 3,000 reviews
- Testing data: 3,000 reviews

A random state of 42 was used so that the same sample could be reproduced when the notebook was run again.

6 Dataset Features

The dataset originally contained the following columns:

- **id:** A unique identifier assigned to each review
- **text:** The written customer review
- **label:** The numerical rating label
- **label_text:** The text representation of the rating label

The text column was renamed to review to make its purpose clearer in the notebook. The original five rating labels were converted into three sentiment classes:

- Labels 0 and 1 were classified as Negative
- Label 2 was classified as Neutral
- Labels 3 and 4 were classified as Positive

7 Data Cleaning

The dataset was inspected and cleaned before preparing it for machine learning. The following cleaning steps were performed:

- Missing review text and labels were checked
- Duplicate reviews were identified and removed
- Review text was converted to lowercase
- HTML tags were removed
- URLs were removed
- Unnecessary characters were removed
- Extra spaces were removed
- Empty reviews were removed after cleaning
- Review length was calculated using word count

Duplicate reviews were also checked across the training, validation, and testing datasets. Reviews appearing in more than one split were removed to reduce the risk of data leakage.

7.1 Cleaning Results

The selected working sample originally contained 30,000 training reviews, 3,000 validation reviews, and 3,000 testing reviews. The training dataset did not contain any missing values or completely duplicated rows. However, 40 duplicate review texts were found in the training data. After removing the duplicate review texts, the training dataset contained 29,960 reviews. The validation dataset contained 2,999 reviews after its initial cleaning. Reviews that were already present in the training dataset were then removed, leaving 2,992 validation reviews. The testing dataset contained 3,000 reviews after its initial cleaning. Reviews that were already present in the training or validation datasets were then removed, leaving 2,990 testing reviews. The final cleaned dataset contained:

- Training reviews: 29,960
- Validation reviews: 2,992
- Testing reviews: 2,990

- Total cleaned reviews: 35,942

Removing repeated reviews across the dataset splits reduced the risk of data leakage. This will help ensure that future machine-learning models are evaluated using reviews that were not already present in the training data.

8 Outlier Handling

Review length was examined using the number of words in each review. Very short and very long reviews were identified using the review-length distribution. Long reviews were not automatically removed because they may contain useful and detailed customer feedback. Review-length analysis was mainly used to understand the dataset and identify unusual records. Review length was treated as a supporting feature rather than direct evidence of sentiment. A long review is not necessarily negative, and a short review is not necessarily positive.

9 Feature Strategy

Machine-learning models cannot directly process normal text. Therefore, the cleaned review text was converted into numerical features using TF-IDF. TF-IDF gives importance to words that are useful in a review but are not equally common across all reviews. The vectorizer included both individual words and two-word combinations. A maximum of 10,000 TF-IDF features was used. The vectorizer was fitted only on the training data. The fitted vectorizer was then used to transform the validation and testing data. This prevented information from the validation and testing datasets from entering the training process. Separate standard scaling was not applied because TF-IDF already converts text into weighted numerical features.

9.1 Feature Extraction Results

After cleaning, TF-IDF was used to convert the customer reviews into numerical feature matrices. The resulting TF-IDF matrix sizes were:

- Training matrix: (29,960, 10,000)
- Validation matrix: (2,992, 10,000)
- Testing matrix: (2,990, 10,000)

In each matrix shape, the first value represents the number of customer reviews, while the second value represents the number of TF-IDF text features. The TF-IDF vectorizer was fitted only on the 29,960 training reviews. The same fitted vectorizer was used to transform the validation and testing reviews. This helped prevent data leakage during feature preparation.

10 Exploratory Data Analysis

Exploratory data analysis was performed to understand the quality and distribution of the customer review dataset. The following areas were analyzed:

- Distribution of positive, neutral, and negative reviews
- Distribution of the original rating labels
- Distribution of review lengths
- Comparison of review lengths between sentiment classes

10.1 Sentiment Distribution

Figure 1 shows the distribution of positive, neutral, and negative reviews in the final cleaned dataset.

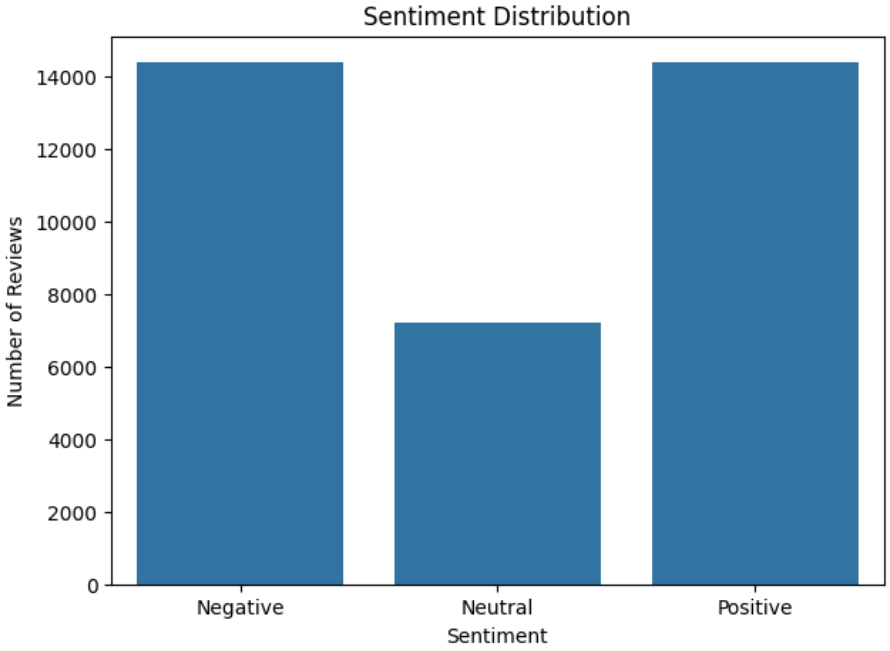


Figure 1: Distribution of sentiment classes

The final cleaned dataset contained 14,376 negative reviews, 14,374 positive reviews, and 7,192 neutral reviews. The negative and positive classes contained approximately twice as many reviews as the neutral class. This occurred because two original rating labels were converted to negative sentiment, one rating label was converted to neutral sentiment, and two rating labels were converted to positive sentiment.

10.2 Rating Distribution

Figure 2 shows the distribution of the five original rating labels.

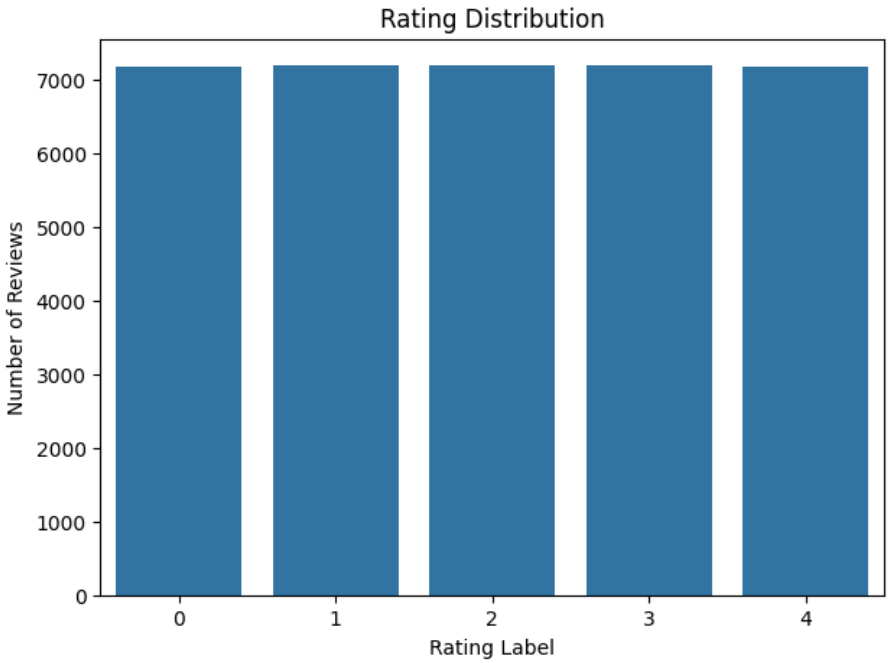


Figure 2: Distribution of original rating labels

The original sampled training dataset contained 6,000 reviews from each of the five rating classes. The rating classes were equally represented because stratified sampling was used when the smaller training sample was created. After the five rating classes were combined into three sentiment categories, the positive and negative classes became larger than the neutral class. This happened because positive and negative sentiment each contained two original rating categories, while neutral sentiment contained only one.

10.3 Review Length Distribution

Figure 3 shows the distribution of review lengths based on word count.

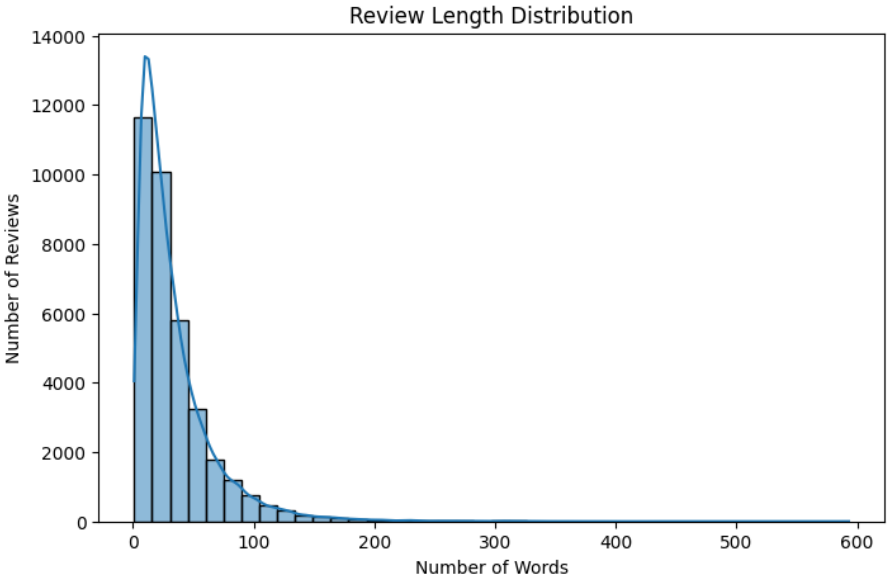


Figure 3: Distribution of review lengths

Most customer reviews were relatively short, while fewer reviews contained a large number of words. Longer reviews were kept because they may contain detailed information about customer experiences, product problems, and complaints.

10.4 Review Length by Sentiment

Figure 4 compares review lengths across the three sentiment classes.

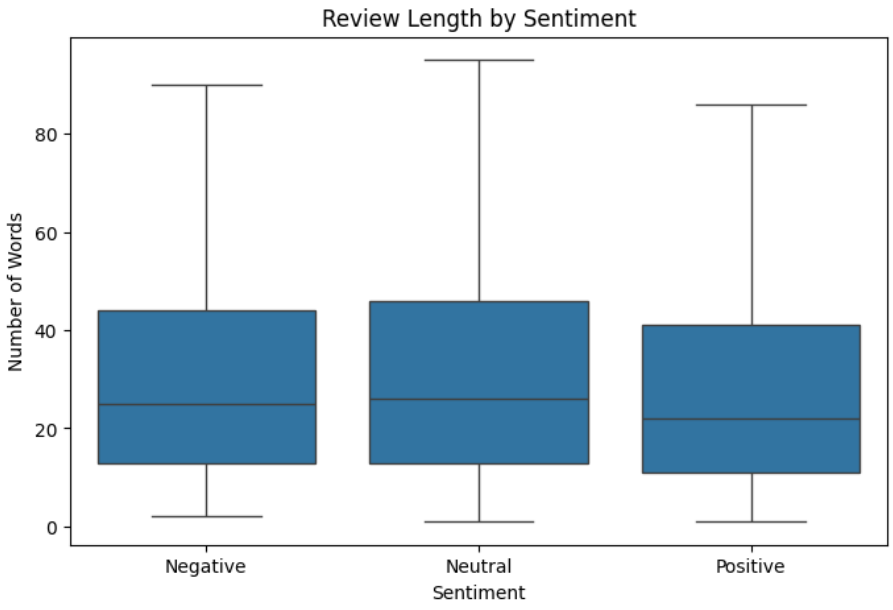


Figure 4: Review length by sentiment class

This comparison was used to examine whether positive, neutral, and negative reviews differed in length. Review length alone was not used to determine sentiment, but it helped in understanding the structure of the dataset.

11 Ethical Considerations and Limitations

Customer review datasets may contain informal writing, spelling mistakes, class imbalance, sarcasm, and mixed opinions. A model trained using reviews from one source may not perform equally well on reviews from another platform. The model may also misunderstand reviews with limited context or opinions expressed indirectly. For example, sarcasm may use positive words to communicate a negative opinion, making it difficult for a basic sentiment model to classify correctly. The sentiment classes were created using ratings. This assumes that the sentiment expressed in the review text agrees with the rating given by the customer. In some cases, the written review and rating may not completely match. The project will not intentionally use personally identifiable customer information. The review ID will only be used as a record identifier and will not be used as a predictive feature.

12 Future Work

The next stages of the project will include:

- Training different sentiment-classification models
- Comparing models using accuracy, precision, recall, and F1-score
- Selecting the best-performing model
- Identifying product and service aspects in customer reviews
- Detecting recurring customer complaints
- Generating recommendations for businesses
- Developing a web application for review analysis

Possible machine-learning models for sentiment classification include Naive Bayes, Logistic Regression, and Support Vector Machine. These models will be trained using the TF-IDF features prepared during this deliverable.

13 Conclusion

This deliverable defined the problem addressed by ReviewIQ AI and prepared the customer review dataset for future machine-learning development. The dataset was acquired using a reproducible method through the Hugging Face datasets library. A stratified working sample was selected, and the review text was checked for missing values, duplicates, and unusual review lengths. After cleaning and cross-split duplicate removal, the final dataset contained 35,942 customer reviews. The cleaned reviews were converted into TF-IDF feature matrices containing a maximum of 10,000 text features. The next phase will focus on training and evaluating machine-learning models for sentiment classification. Aspect detection, recurring complaint identification, and business recommendations will then be added to develop the complete ReviewIQ AI application.

References

[1] SetFit. *Amazon Reviews Multi English Dataset*. Hugging Face.

[2] Hugging Face. *Datasets Library*.

[3] Scikit-learn Developers. *Scikit-learn Machine Learning Library*.